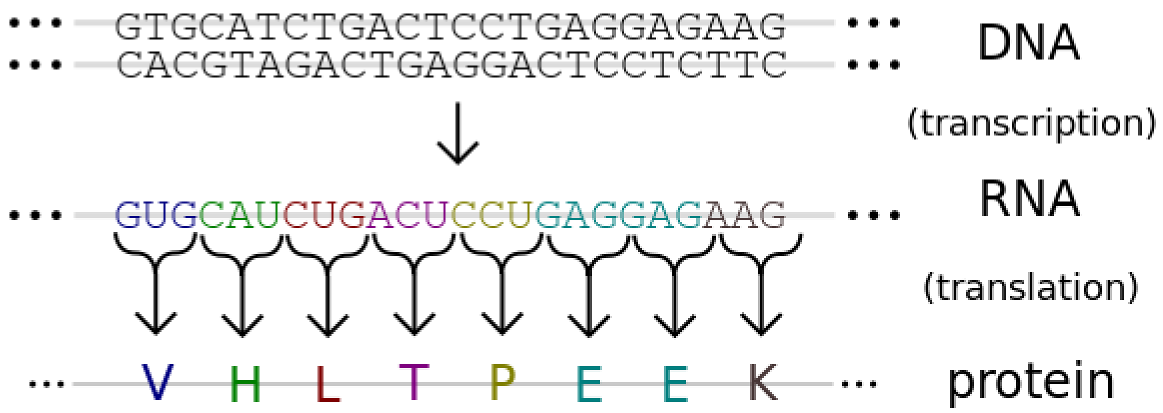
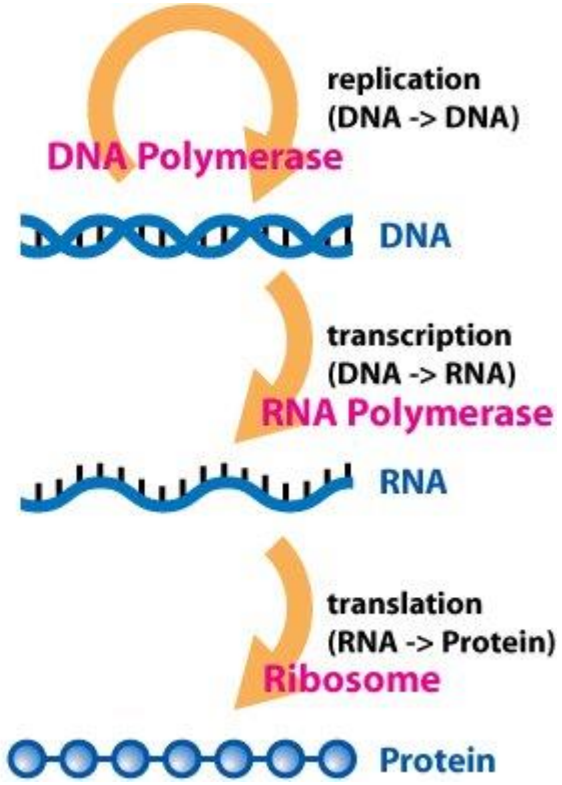
中心法则



GTGCATCTGACTCCTGAGGAGAAG
CACGTAGACTGAGGACTCCTCTTC
DNA
(transcription)
GUGCAUCUGACUCCUGAGGAGAAG
RNA
(translation)
V H L T P E E K
protein



replication
(DNA -> DNA)
DNA Polymerase
DNA
transcription
(DNA -> RNA)
RNA Polymerase
RNA
translation
(RNA -> Protein)
Ribosome
Protein

本堂关注重点：蛋白质数据库

- Uniprot：序列＋注释
- PDB：　　3D 结构

The Universal Protein Resource

（UniProt）

http://www.uniprot.org/

内容提纲

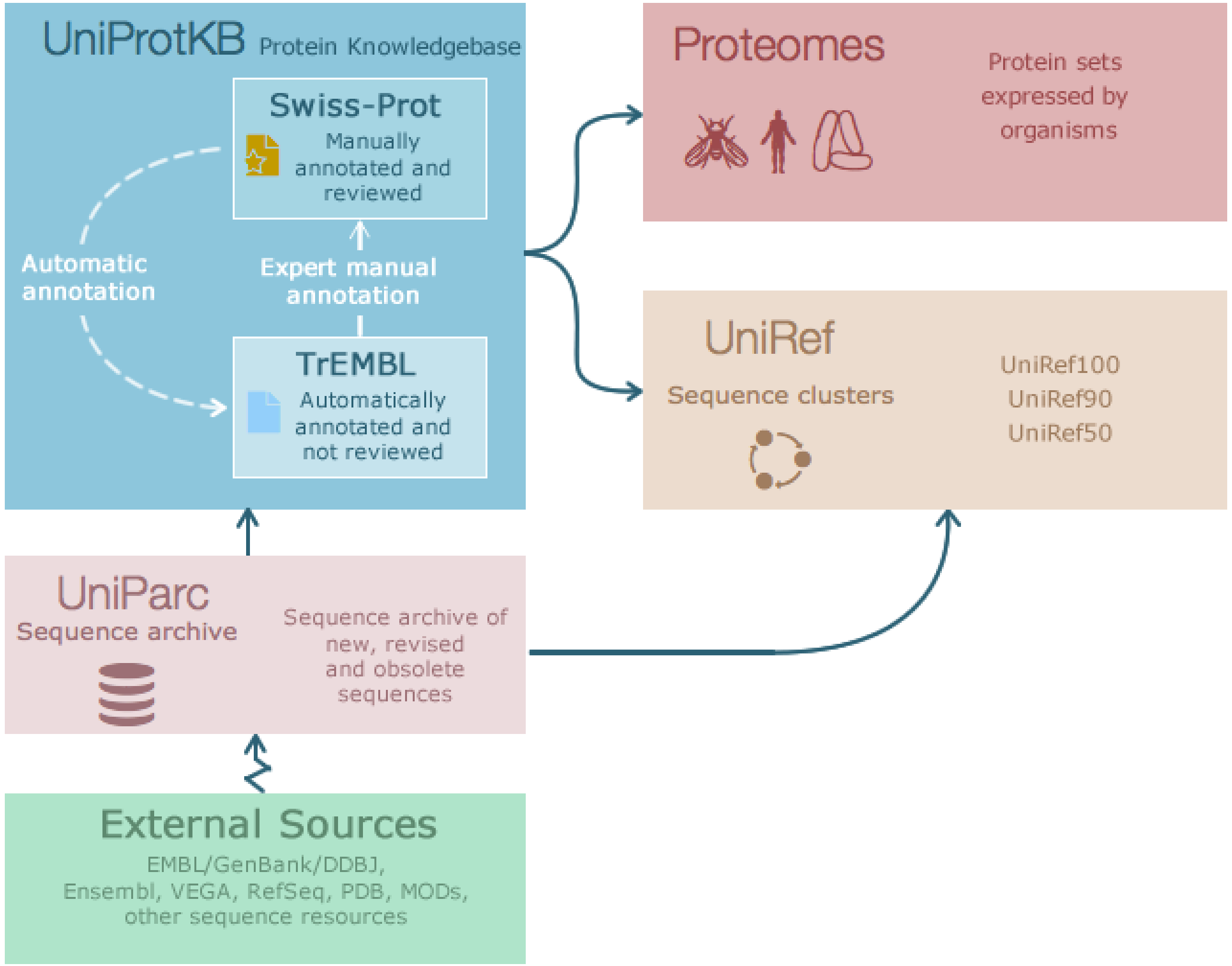
- UniProt 介绍
- 如何检索
- 如何下载

Uniprot简介

- 全称：The Universal Protein Resource
- The mission of UniProt is to provide the scientific community with a comprehensive, high-quality and freely accessible resource of protein sequence and functional information.
- 合作单位：
  - the European Bioinformatics Institute (EMBL-EBI),
  - the SIB Swiss Institute of Bioinformatics
  - the Protein Information Resource (PIR)

Uniprot简介

- 包含：
  1) UniProt Knowledgebase (UniProtKB),
  2) UniProt Reference Clusters (UniRef),
  3) UniProt Archive (UniParc)



1) The UniProt Knowledgebase (UniProtKB)

- The centerpiece of the UniProt Consortium's activity

- 包含：
  - UniProtKB/Swiss-Prot:

    contains high-quality manually annotated and non-redundant protein sequence records
  - UniProtKB/TrEMBL

    contains high-quality computationally analyzed records enriched with automatic annotation and classification

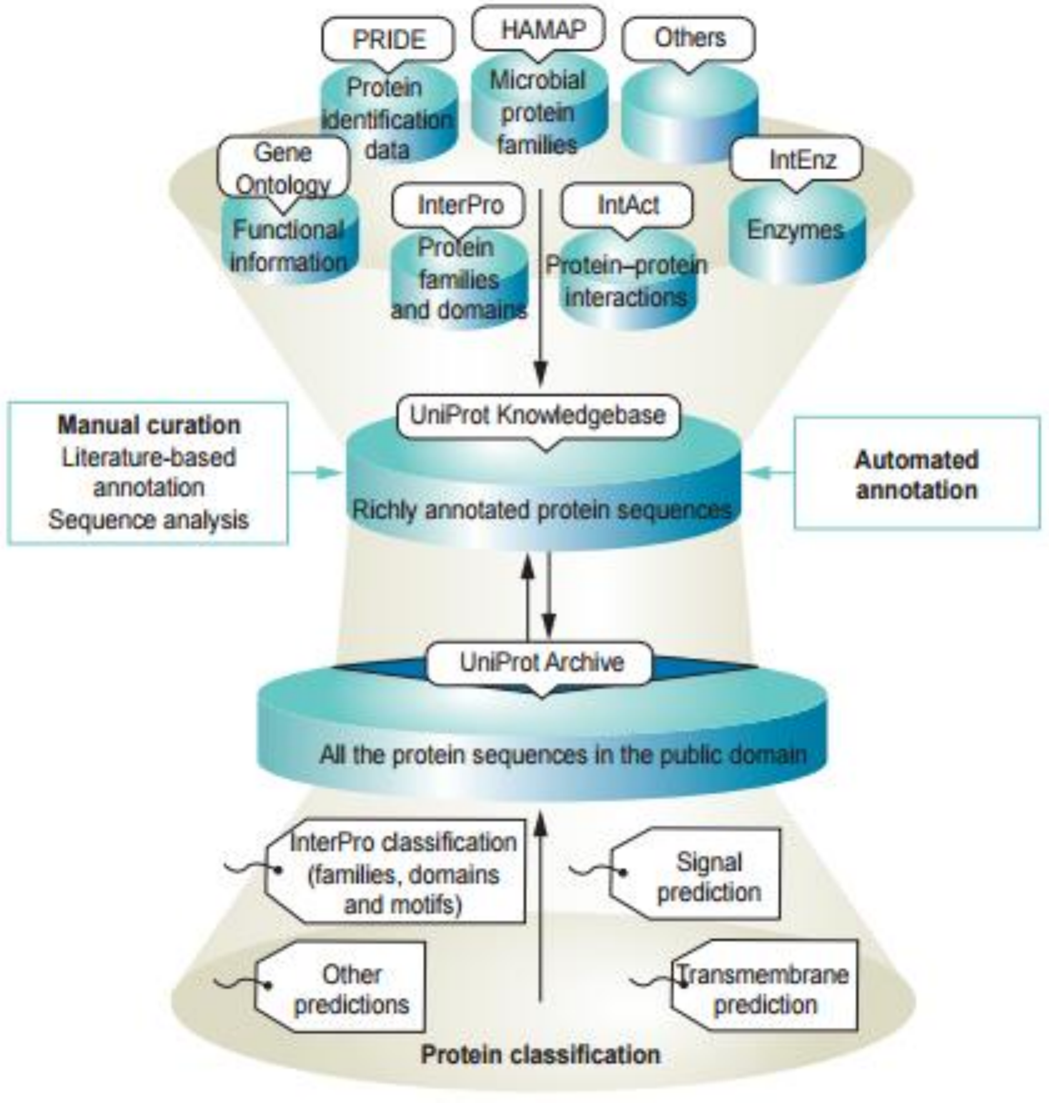
- 序列来源：
  - DDBJ/ENA/GenBank coding sequence (CDS) translations
  - the sequences of PDB structures
  - sequences from Ensembl and RefSeq
  - data derived from amino acid sequences that are directly submitted to UniProtKB or scanned from the literature.

2) UniProt Reference Clusters (UniRef)

- Three UniRef databases – UniRef100, UniRef90 and UniRef50 – merge sequences automatically across species.
- chicken, cow, dog, fly, Fugu, human, mouse, rat, Tetraodon, Xenopus and zebrafish
- UniRef90 and UniRef50 are built from UniRef100 to provide records with mutual sequence identity of 90% or more, or 50% or more, respectively, with links to the corresponding UniProtKB records.

3) UniProt Archive (UniParc)

- UniParc is designed to capture all publicly available protein sequence data and contains all the protein sequences from the main publicly available protein sequence databases
- 大型容纳器



Sources of annotation for the UniProt Knowledgebase.

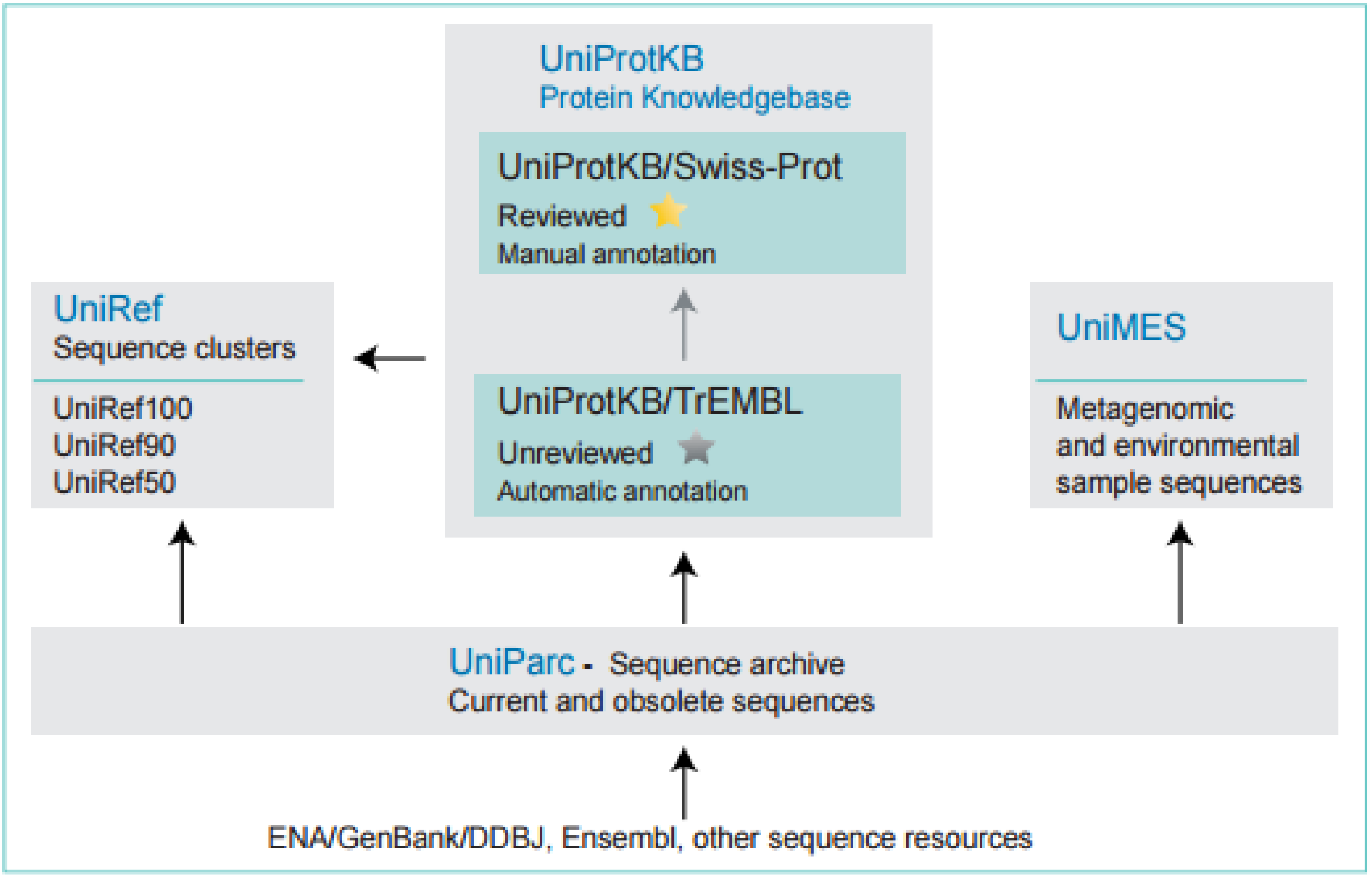
Different databases for different uses

- The UniProt Knowledgebase → access functional information on proteins
  - Each entry contains:
    - The amino acid sequence
    - Protein name or description
    - Taxonomic data
    - Citation information
    - Biological ontology, classification, and cross-references
    - ….

Different databases for different uses

- The UniRef databases
  - → provide clustered sets of sequences (依据序列相似度)
  - → provide complete coverage of sequence space at several resolutions
- UniParc :
  - → the most comprehensive publicly accessible non-redundant protein sequence database available, providing links to all underlying sources and versions of these sequences

UniProt 数据库结构回顾



内容提纲

- UniProt 介绍
- 如何检索
- 如何下载

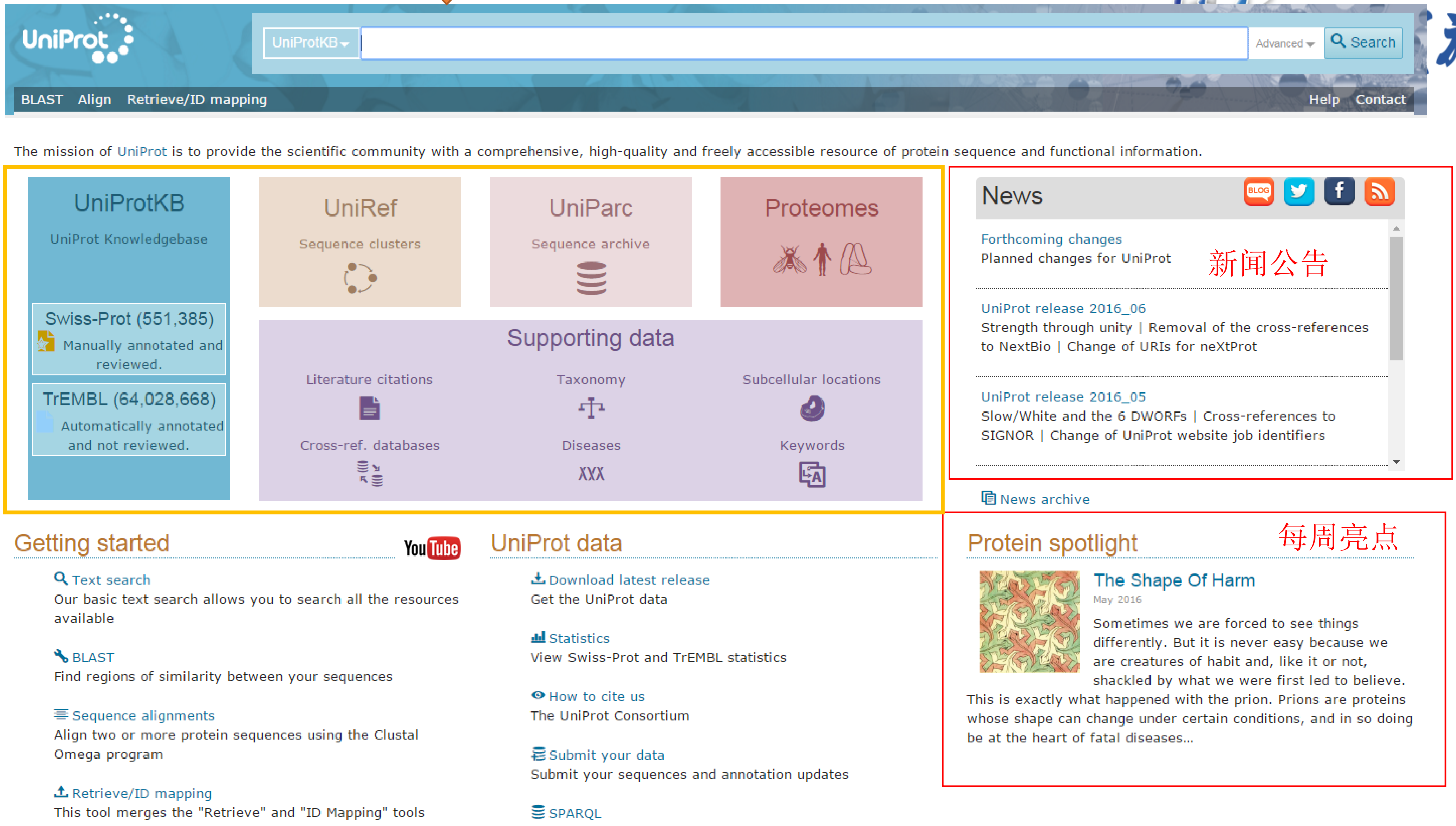
看起来结果复杂的数据库，使用起来是否也同样复杂呢？

NO！！！

http://www.uniprot.org/

检索框

详细的分栏

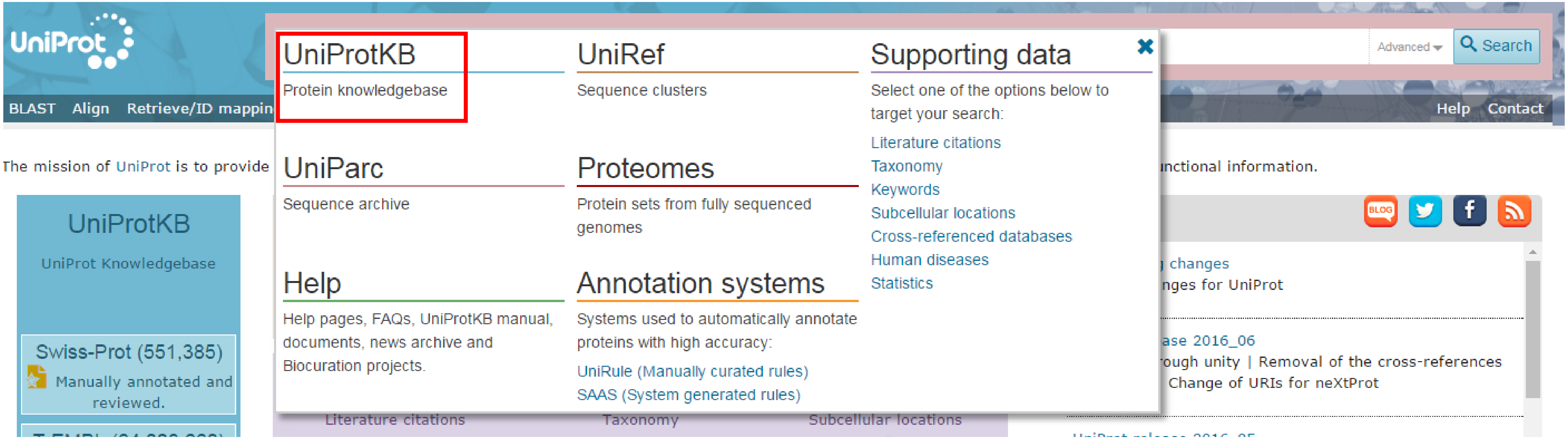


检索方式

1. 文字检索
2. 序列相似形检索 （Blast）

通过文字检索

- 1）选定要采用的数据库



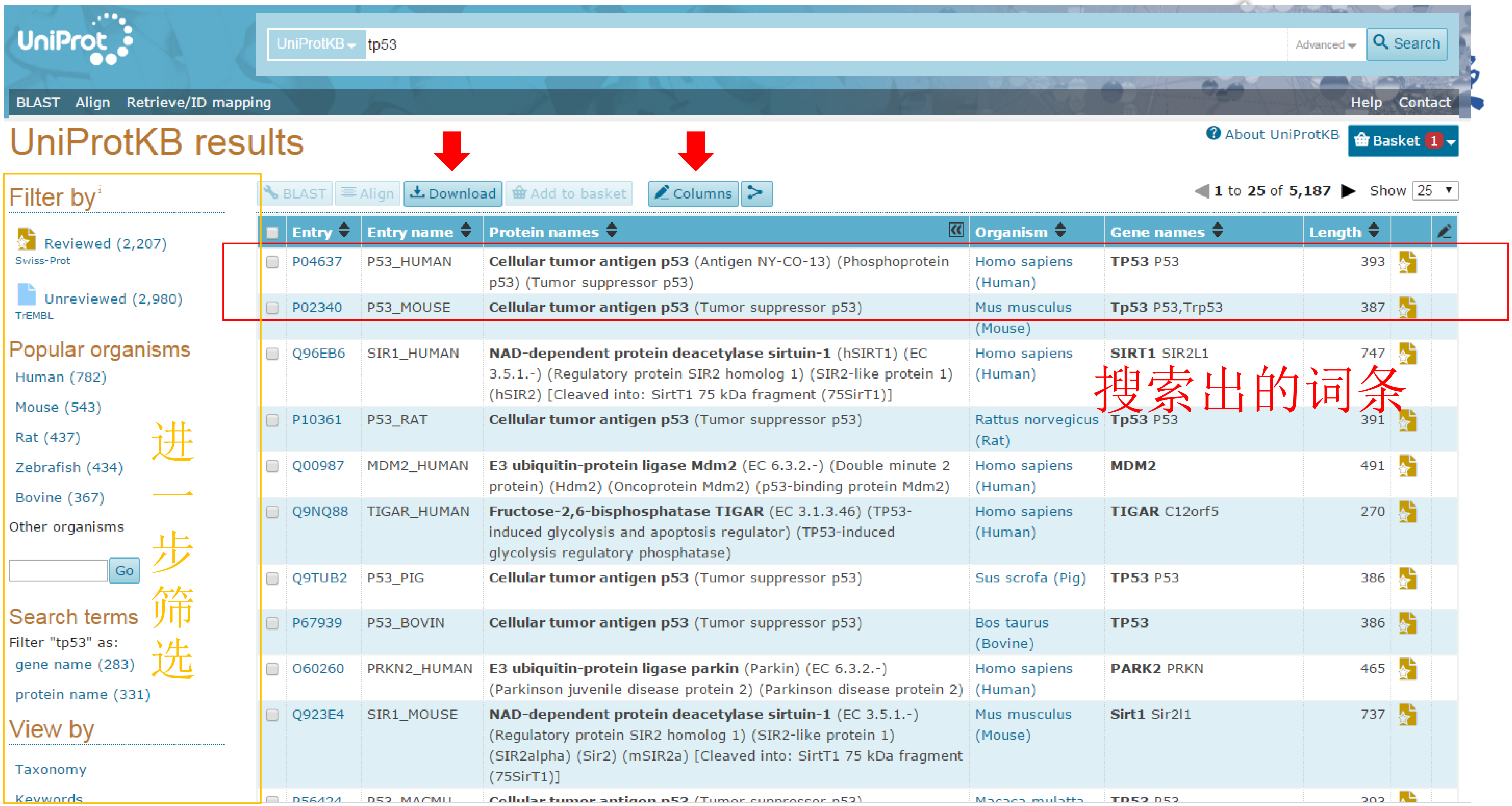
当选定了不同的子数据库后，搜索条的背景颜色会相应的做出改变，以提醒用户。

通过文字检索

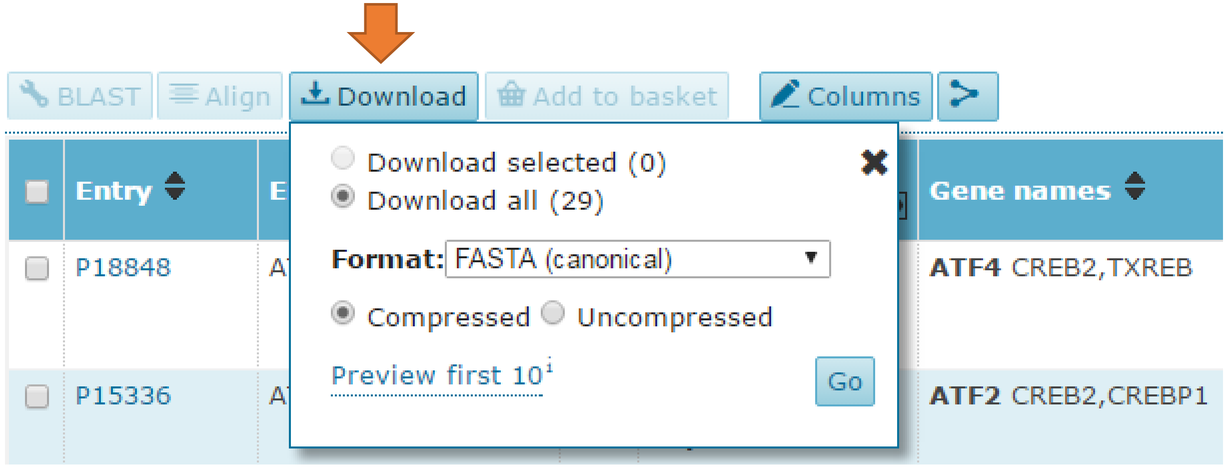
2) 键入关键词

文字检索格式

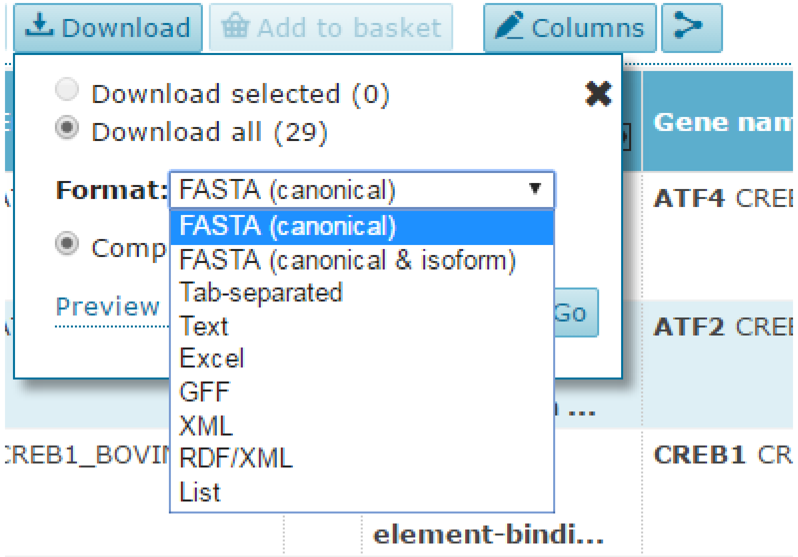
human antigen	All entries containing both terms.
human AND antigen	
human && antigen	
"human antigen"	All entries containing both terms in the exact order.
human -antigen	All entries containing the term human but not antigen.
human NOT antigen	
human ! antigen	
human OR mouse	All entries containing either term.
human || mouse	
antigen AND (human OR mouse)	Using parentheses to override boolean precedence rules.
anti*	All entries containing terms starting with anti. Asterisks can also be used at the beginning and within terms. **Note:** Terms starting with an asterisk or a single letter followed by an asterisk can slow down queries considerably.
author:Tiger*	Citations that have an author whose name starts with Tiger. To search in a specific field of a dataset, you must prefix your search term with the field name and a colon. To discover what fields can be queried explicitly, observe the query hints that are shown after submitting a query or use the query builder (see below).
length:[100 TO *]	All entries with a sequence of at least 100 amino acids.
citation:(author:Arai author:Chung)	All entries with a publication that was coauthored by two specific authors


UniProt
UniProtKB
tp53
Advanced
Search
BLAST
Align
Retrieve/ID mapping
Help
Contact
UniProtKB results
About UniProtKB
Basket 1
Filter by
Reviewed (2,207)
Swiss-Prot
Unreviewed (2,980)
TrEMBL
Popular organisms
Human (782)
Mouse (543)
Rat (437)
Zebrafish (434)
Bovine (367)
Other organisms
Go
Search terms
Filter "tp53" as:
gene name (283)
protein name (331)
View by
Taxonomy
Keywords
进一步筛选
BLAST
Align
Download
Add to basket
Columns
1 to 25 of 5,187
Show 25
Entry | Entry name | Protein names | Organism | Gene names | Length
P04637 | P53_HUMAN | Cellular tumor antigen p53 (Antigen NY-CO-13) (Phosphoprotein p53) (Tumor suppressor p53) | Homo sapiens (Human) | TP53 P53 | 393
P02340 | P53_MOUSE | Cellular tumor antigen p53 (Tumor suppressor p53) | Mus musculus (Mouse) | Tp53 P53,Trp53 | 387
Q96EB6 | SIR1_HUMAN | NAD-dependent protein deacetylase sirtuin-1 (hSIRT1) (EC 3.5.1.-) (Regulatory protein SIR2 homolog 1) (SIR2-like protein 1) (hSIR2) [Cleaved into: SirtT1 75 kDa fragment (75SirT1)] | Homo sapiens (Human) | SIRT1 SIR2L1 | 747
P10361 | P53_RAT | Cellular tumor antigen p53 (Tumor suppressor p53) | Rattus norvegicus (Rat) | Tp53 P53 | 391
Q00987 | MDM2_HUMAN | E3 ubiquitin-protein ligase Mdm2 (EC 6.3.2.-) (Double minute 2 protein) (Hdm2) (Oncoprotein Mdm2) (p53-binding protein Mdm2) | Homo sapiens (Human) | MDM2 | 491
Q9NQ88 | TIGAR_HUMAN | Fructose-2,6-bisphosphatase TIGAR (EC 3.1.3.46) (TP53-induced glycolysis and apoptosis regulator) (TP53-induced glycolysis regulatory phosphatase) | Homo sapiens (Human) | TIGAR C12orf5 | 270
Q9TUB2 | P53_PIG | Cellular tumor antigen p53 (Tumor suppressor p53) | Sus scrofa (Pig) | TP53 P53 | 386
P67939 | P53_BOVIN | Cellular tumor antigen p53 (Tumor suppressor p53) | Bos taurus (Bovine) | TP53 | 386
O60260 | PRKN2_HUMAN | E3 ubiquitin-protein ligase parkin (Parkin) (EC 6.3.2.-) (Parkinson juvenile disease protein 2) (Parkinson disease protein 2) | Homo sapiens (Human) | PARK2 PRKN | 465
Q923E4 | SIR1_MOUSE | NAD-dependent protein deacetylase sirtuin-1 (EC 3.5.1.-) (Regulatory protein SIR2 homolog 1) (SIR2-like protein 1) (SIR2alpha) (Sir2) (mSIR2a) [Cleaved into: SirtT1 75 kDa fragment (75SirT1)] | Mus musculus (Mouse) | Sirt1 Sir2l1 | 737
搜索出的词条

- 下载搜索结果列表



可用的格式



• 根据自己的需要，定制搜索结果列表的显示方式

Did you mean tp53? 5,187 result(s) found
Expand search "Homo sapiens (Human) [9606]" to include lower taxonomic ranks

Entry	Entry name	Protein names	Organism	Gene names	Length	
P04637	P53_HUMAN	**Cellular tumor antigen p53** (Antigen NY-CO-13) (Phosphoprotein p53) (Tumor suppressor p53)	Homo sapiens (Human)	**TP53** P53	393	
Q96EB6	SIR1_HUMAN	**NAD-dependent protein deacetylase sirtuin-1** (hSIRT1) (EC 3.5.1.-) (Regulatory protein SIR2 homolog 1) (SIR2-like protein 1) (hSIR2) [Cleaved into: SirtT1 75 kDa fragment (75SirT1)]	Homo sapiens (Human)	**SIRT1** SIR2L1	747	
Q00987	MDM2_HUMAN	**E3 ubiquitin-protein ligase Mdm2** (EC 6.3.2.-) (Double minute 2 protein) (Hdm2) (Oncoprotein Mdm2) (p53-binding protein Mdm2)	Homo sapiens (Human)	**MDM2**	491	
Q9NQ88	TIGAR_HUMAN	**Fructose-2,6-bisphosphatase TIGAR** (EC 3.1.3.46) (TP53-induced glycolysis and apoptosis regulator) (TP53-induced glycolysis regulatory phosphatase)	Homo sapiens (Human)	**TIGAR** C12orf5	270	

UniProt 编号

UniProt 编号+物种缩写

蛋白名

物种

基因名

长度

来源

不同的信息显示方式

UniProtKB - P04637 (P53_HUMAN)

Basket

Display

Entry
Feature viewer
Feature table

All None

- Function
- Names & Taxonomy
- Subcellular location
- Pathology & Biotech
- PTM / Processing
- Expression
- Interaction
- Structure
- Family & Domains
- Sequences (9)
- Cross-references
- Publications
- Entry information
- Miscellaneous
- Similar proteins

Top

各类详细的注释信息

BLAST Align Format Add to basket History

Feedback Help video Other tutorials and videos

Protein Cellular tumor antigen p53

Gene TP53

Organism *Homo sapiens (Human)*

Status Reviewed - Annotation score: - Experimental evidence at protein level

Function

Acts as a tumor suppressor in many tumor types; induces growth arrest or apoptosis depending on the physiological circumstances and cell type. Involved in cell cycle regulation as a trans-activator that acts to negatively regulate cell division by controlling a set of genes required for this process. One of the activated genes is an inhibitor of cyclin-dependent kinases. Apoptosis induction seems to be mediated either by stimulation of BAX and FAS antigen expression, or by repression of Bcl-2 expression. In cooperation with mitochondrial PPIF is involved in activating oxidative stress-induced necrosis; the function is largely independent of transcription. Induces the transcription of long intergenic non-coding RNA p21 (lincRNA-p21) and lincRNA-Mkln1. LincRNA-p21 participates in TP53-dependent transcriptional repression leading to apoptosis and seem to have to effect on cell-cycle regulation. Implicated in Notch signaling cross-over. Prevents CDK7 kinase activity when associated to CAK complex in response to DNA damage, thus stopping cell cycle progression. Isoform 2 enhances the transactivation activity of isoform 1 from some but not all TP53-inducible promoters. Isoform 4 suppresses transactivation activity and impairs growth suppression mediated by isoform 1. Isoform 7 inhibits isoform 1-mediated apoptosis. Regulates the circadian clock by repressing CLOCK-ARNTL/BMAL1-mediated transcriptional activation of PER2 (PubMed:24051492). 12 Publications

Cofactor

Zn^{2+}

Note: Binds 1 zinc ion per subunit.

Sites

Feature key	Position(s)	Length	Description	Graphical view	Feature identifier	Actions
Metal binding	176 – 176	1	Zinc			
Metal binding	179 – 179	1	Zinc			
Metal binding	238 – 238	1	Zinc			
Metal binding	242 – 242	1	Zinc			

功能描述

Protein **Cellular tumor antigen p53**

Gene **TP53**

Organism *Homo sapiens (Human)*

Status Reviewed - Annotation score: - Experimental evidence at protein level[i]

Function[i]

Acts as a tumor suppressor in many tumor types; induces growth arrest or apoptosis depending on the physiological circumstances and cell type. Involved in cell cycle regulation as a trans-activator that acts to negatively regulate cell division by controlling a set of genes required for this process. One of the activated genes is an inhibitor of cyclin-dependent kinases. Apoptosis induction seems to be mediated either by stimulation of BAX and FAS antigen expression, or by repression of Bcl-2 expression. In cooperation with mitochondrial PPIF is involved in activating oxidative stress-induced necrosis; the function is largely independent of transcription. Induces the transcription of long intergenic non-coding RNA p21 (lincRNA-p21) and lincRNA-Mkln1. LincRNA-p21 participates in TP53-dependent transcriptional repression leading to apoptosis and seem to have to effect on cell-cycle regulation. Implicated in Notch signaling cross-over. Prevents CDK7 kinase activity when associated to CAK complex in response to DNA damage, thus stopping cell cycle progression. Isoform 2 enhances the transactivation activity of isoform 1 from some but not all TP53-inducible promoters. Isoform 4 suppresses transactivation activity and impairs growth suppression mediated by isoform 1. Isoform 7 inhibits isoform 1-mediated apoptosis. Regulates the circadian clock by repressing CLOCK-ARNTL/BMAL1-mediated transcriptional activation of PER2 (PubMed:24051492). 12 Publications

Manual assertion based on experiment in[i]

"Regulation of CAK kinase activity by p53."
Schneider E., Montenarh M., Wagner P.
Oncogene 17:2733-2741(1998) [PubMed] [Europe PMC] [Abstract]
Cited for: FUNCTION, IDENTIFICATION IN COMPLEX WITH CAK.

Cofactor[i]

Zn^{2+}

Note: Binds 1 zinc ion per subunit.

Sites

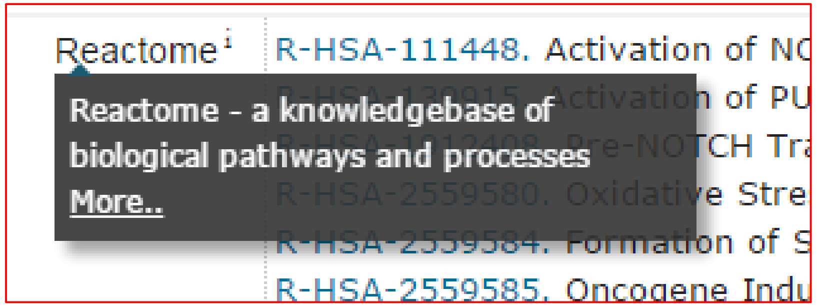
Feature key	Position(s)	Length	Description	...ifier	Actions
Metal binding[i]	176 – 176	1	Zinc		
Metal binding[i]	179 – 179	1	Zinc		

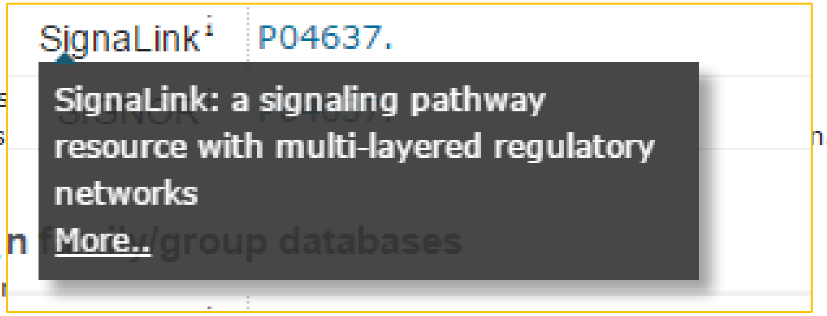
GO - Molecular function [i]

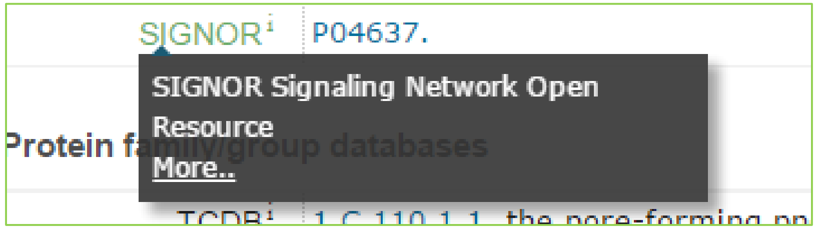
- ATP binding Source: UniProtKB
- chaperone binding Source: UniProtKB
- chromatin binding Source: UniProtKB
- copper ion binding Source: UniProtKB
- core promoter sequence-specific DNA binding Source: UniProtKB
- damaged DNA binding Source: GO_Central
- DNA binding Source: UniProtKB
- double-stranded DNA binding Source: GO_Central
- enzyme binding Source: UniProtKB
- histone acetyltransferase binding Source: UniProtKB
- identical protein binding Source: IntAct
- p53 binding Source: GO_Central
- protease binding Source: UniProtKB
- protein heterodimerization activity Source: UniProtKB
- protein kinase binding Source: UniProtKB
- protein N-terminus binding Source: UniProtKB
- protein phosphatase 2A binding Source: UniProtKB
- protein phosphatase binding Source: UniProtKB
- protein self-association Source: AgBase
- receptor tyrosine kinase binding Source: BHF-UCL
- RNA polymerase II transcription factor activity, sequence-specific DNA binding Source: UniProtKB
- RNA polymerase II transcription factor binding Source: BHF-UCL

Enzyme and pathway databases

Reactome[i]	R-HSA-111448. Activation of NOXA and translocation to mitochondria. R-HSA-139915. Activation of PUMA and translocation to mitochondria. R-HSA-1912408. Pre-NOTCH Transcription and Translation. R-HSA-2559580. Oxidative Stress Induced Senescence. R-HSA-2559584. Formation of Senescence-Associated Heterochromatin Foci (SAHF). R-HSA-2559585. Oncogene Induced Senescence. R-HSA-2559586. DNA Damage/Telomere Stress Induced Senescence. R-HSA-3232118. SUMOylation of transcription factors. R-HSA-349425. Autodegradation of the E3 ubiquitin ligase COP1. R-HSA-390471. Association of TriC/CCT with target proteins during biosynthesis. R-HSA-5628897. TP53 Regulates Metabolic Genes. R-HSA-5693565. Recruitment and ATM-mediated phosphorylation of repair and signaling R-HSA-6796648. TP53 Regulates Transcription of DNA Repair Genes. R-HSA-6803204. TP53 Regulates Transcription of Genes Involved in Cytochrome C Releas R-HSA-6803205. TP53 regulates transcription of several additional cell death genes whos R-HSA-6803207. TP53 Regulates Transcription of Caspase Activators and Caspases. R-HSA-6803211. TP53 Regulates Transcription of Death Receptors and Ligands. R-HSA-6804114. TP53 Regulates Transcription of Genes Involved in G2 Cell Cycle Arrest. R-HSA-6804115. TP53 regulates transcription of additional cell cycle genes whose exact r R-HSA-6804116. TP53 Regulates Transcription of Genes Involved in G1 Cell Cycle Arrest. R-HSA-6804754. Regulation of TP53 Expression. R-HSA-6804756. Regulation of TP53 Activity through Phosphorylation. R-HSA-6804757. Regulation of TP53 Degradation. R-HSA-6804758. Regulation of TP53 Activity through Acetylation. R-HSA-6804759. Regulation of TP53 Activity through Association with Co-factors. R-HSA-6804760. Regulation of TP53 Activity through Methylation. R-HSA-69473. G2/M DNA damage checkpoint. R-HSA-69481. G2/M Checkpoints. R-HSA-69541. Stabilization of p53. R-HSA-69895. Transcriptional activation of cell cycle inhibitor p21. R-HSA-8852276. The role of GTSE1 in G2/M progression after G2 checkpoint. R-HSA-983231. Factors involved in megakaryocyte development and platelet production.
SignaLink[i]	P04637.
SIGNOR[i]	P04637.







Pathology & Biotech[i]

Involvement in disease[i]

TP53 is found in increased amounts in a wide variety of transformed cells. TP53 is frequently mutated or inactivated in about 60% of cancers. TP53 defects are found in Barrett metaplasia a condition in which the normally stratified squamous epithelium of the lower esophagus is replaced by a metaplastic columnar epithelium. The condition develops as a complication in approximately 10% of patients with chronic gastroesophageal reflux disease and predisposes to the development of esophageal adenocarcinoma.

Esophageal cancer (ESCR)

The disease is caused by mutations affecting the gene represented in this entry.
Disease description: A malignancy of the esophagus. The most common types are esophageal squamous cell carcinoma and adenocarcinor
remains a devastating disease because it is usually not detected until it has progressed to an advanced incurable stage.
See also OMIM:133239

Li-Fraumeni syndrome (LFS) 9 Publications

The disease is caused by mutations affecting the gene represented in this entry.
Disease description: Autosomal dominant familial cancer syndrome that in its classic form is defined by the existence of a proband affecte rs
with a first degree relative affected by any tumor before 45 years and another first degree relative with any tumor before 45 years or a s cal
definitions for LFS have been proposed (PubMed:8118819 and PubMed:8718514) and called Li-Fraumeni like syndrome (LFL). In these fami a
diverse set of malignancies at unusually early ages. Four types of cancers account for 80% of tumors occurring in TP53 germline mutatior
tissue and bone sarcomas, brain tumors (astrocytomas) and adrenocortical carcinomas. Less frequent tumors include choroid plexus carci
age of 15, rhabdomyosarcoma before the age of 5, leukemia, Wilms tumor, malignant phyllodes tumor, colorectal and gastric cancers.
See also OMIM:151623

Feature key	Position(s)	Length	Description	Graphical view		ns
Natural variant[i]	82 – 82	1	P → L in LFS; germline mutation and in sporadic cancers; somatic mutation.		V	
Natural variant[i]	97 – 97	1	V → I in familial cancer not matching LFS; germline mutation and in a sporadic cancer; somatic mutation.		V	
Natural variant[i]	105 – 105	1	G → C in LFS; germline mutation and in sporadic		VAR_044661	

Function
Names & Taxonomy
Subcellular location
Pathology & Biotech
PTM / Processing
Expression
Interaction
Structure
Family & Domains
Sequences (9)
Cross-references
Publications
Entry information
Miscellaneous
Similar proteins

- Function
- Names & Taxonomy
- Subcellular location
- Pathology & Biotech
- PTM / Processing
- Expression
- Interaction
- Structure
- Family & Domains
- Sequences (9)
- Cross-references
- Publications
- Entry information
- Miscellaneous
- Similar proteins

Structure[1]

Secondary structure

Legend: Helix Turn Beta strand

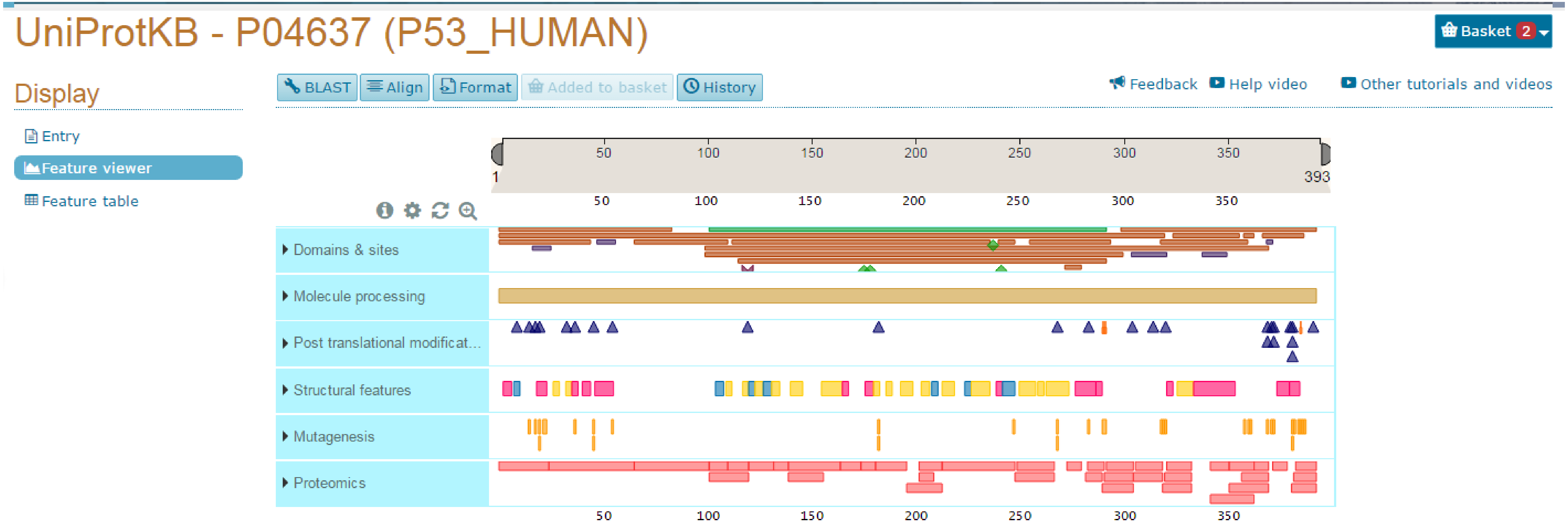
Show more details

3D structure databases

Select the link destinations:
- PDBe
- RCSB PDB
- PDBj

Entry	Method	Resolution (Å)	Chain	Positions	PDBsum
1A1U	NMR	-	A/C	324-358	[»]
1AIE	X-ray	1.50	A	326-356	[»]
1C26	X-ray	1.70	A	325-356	[»]
1DT7	NMR	-	X/Y	367-388	[»]
1GZH	X-ray	2.60	A/C	95-292	[»]
1H26	X-ray	2.24	E	376-386	[»]
1HS5	NMR	-	A/B	324-357	[»]
1JSP	NMR	-	A	367-386	[»]
1KZY	X-ray	2.50	A/B	95-289	[»]

另一类有效的查看方式



- 把冗长的列表和文字信息用图像的方式简单表示。
- 具体每项有可展开

第二类检索方式：Blast

BLAST Align Retrieve/ID mapping Help Contact

BLAST

How to use this tool

The Basic Local Alignment Search Tool (BLAST) finds regions of local similarity between sequences, which can be used to infer functional and evolutionary relationships between sequences as well as help identify members of gene families.

1. Enter either a protein or nucleotide sequence or a UniProt identifier (e.g.P00750 or A4_HUMAN or UPI0000000001) into the form field.
2. Optionally, change the program parameters with the dropdown menus under the form.
3. Click the *Run BLAST* button.

Help | Blast help video | Other tutorials and videos | Downloads

```
>sp|P04637|P53_HUMAN Cellular tumor antigen p53 OS=Homo sapiens GN=TP53 PE=1 SV=4
MEEPQSDPSVEPPLSQETFSDLWKLLPENNVLSPLPSQAMDDLMLSPDDIEQWFTEDPGP
DEAPRMPEAAPPVAPAPAAPTPAAPAPAPSWPLSSSVPSQKTYQGSYGFRLGFLHSGTAK
SVTCTYSPALNKMFCQLAKTCPVQLWVDSTPPPGTRVRAMAIYKQSQHMTEVVRRCPHHE
RCSDSDGLAPPQHLIRVEGNLRVEYLDDRNTFRHSVVVPYEPPEVGSDCTTIHYNYMCNS
SCMGGMNRRPILTIITLEDSSGNLLGRNSFEVRVCACPGRDRRTEEENLRKKGEPHHELP
PGSTKRALPNNTSSSPQPKKKPLDGEYFTLQIRGRERFEMFRELNEALELKDAQAGKEPG
GSRAHSSHLKSKKGQSTSRHKKLMFKTEGPDSD
```

Target database	E-Threshold	Matrix	Filtering	Gapped	Hits
UniProtKB	10	Auto	None	yes	250

Run Blast in a separate window.

Clear | Run BLAST

解螺旋

100 80 60 40 20 0
Identity %

Edit and resubmit Order by: Score

Overview

Show all 236

Entry	Protein names	Match hit (200 400 600)	Identity
H2QC53	Cellular tumor antigen p53 (Pan troglodytes)		100.0%
K7PPA8	Cellular tumor antigen p53 (Homo sapiens)		100.0%
P04637	Cellular tumor antigen p53 (Homo sapiens)		100.0%
G3R2U9	Cellular tumor antigen p53 (Gorilla gorilla gorilla)		99.7%
K7PPU4	Cellular tumor antigen p53 (Homo sapiens)		99.7%

Alignments

BLAST Align Download Add to basket Columns

1 to 25 of 236 Show 25

	Entry	Alignment overview	Info	Status
	Query: sp|P04637|P53_HUMAN B20160612C2335653E4FA1B8AECF5153189FA788F0A77A3C			
	H2QC53	H2QC53_PANTR - Cellular tumor antigen p53 - Pan troglodytes ... - View alignment	E-value: 0.0 Score: 2,121 Ident.: 100.0%	
	K7PPA8	K7PPA8_HUMAN - Cellular tumor antigen p53 Homo sapiens (Human) - View alignment	E-value: 0.0 Score: 2,121 Ident.: 100.0%	
	P04637	P53_HUMAN - Cellular tumor antigen p53 Homo sapiens (Human) - View alignment	E-value: 0.0	

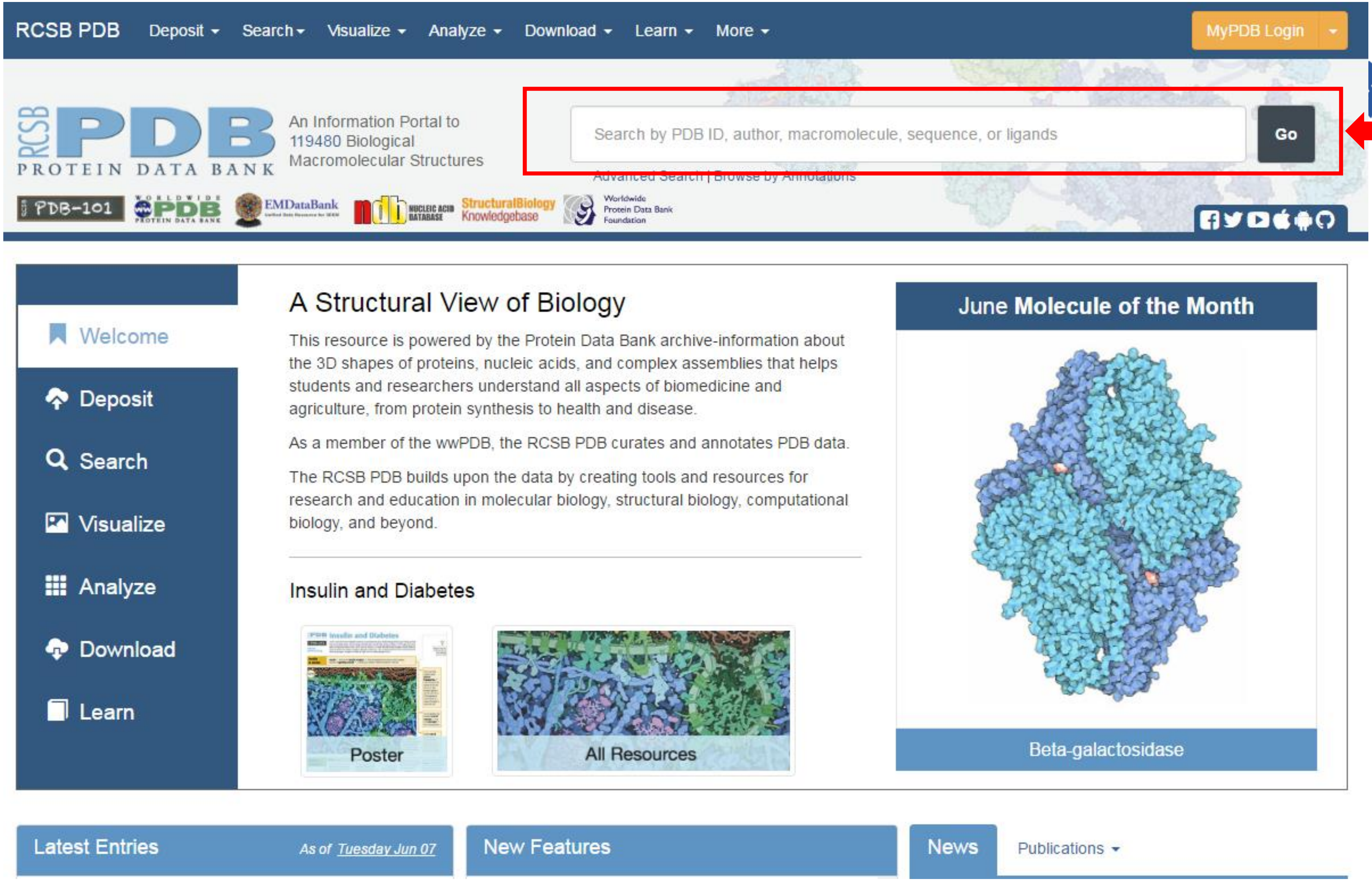
Part2 RCSB PDB PROTEIN DATA BANK

An Information Portal to 119480 Biological Macromolecular Structures

PDB 简介

- An Information Portal to 119480 Biological Macromolecular Structures
- 内容：
  - 3D shapes of proteins, nucleic acids, and complex assemblies that helps students and researchers understand all aspects of biomedicine and agriculture, from protein synthesis to health and disease.
- 数据来源：
  - Structural biologists use methods such as X-ray crystallography, NMR spectroscopy, and cryo-electron microscopy to determine the location of each atom relative to each other in the molecule. They then deposit this information, which is then annotated and publicly released into the archive by the wwPDB.

PDB at a Glance | 38375 Distinct Protein Sequences | 31921 Structures of Human Sequences | 8508 Nucleic Acid Containing Structures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This resource is powered by the Protein Data Bank archive-information about the 3D shapes of proteins, nucleic acids, and complex assemblies that helps students and researchers understand all aspects of biomedicine and agriculture, from protein synthesis to health and disease.
As a member of the wwPDB, the RCSB PDB curates and annotates PDB data.
The RCSB PDB builds upon the data by creating tools and resources for research and education in molecular biology, structural biology, computational biology, and beyond.
Insulin and Diabetes
Poster
All Resources
June Molecule of the Month
Beta-galactosidase
Latest Entries
As of Tuesday Jun 07
New Features
News
Publications

最简单的检索方式

NMR Ensemble

View in 3D JSmol or PV (in Browser)

Standalone Viewers

Simple Viewer   Protein Workshop   Kiosk Viewer

Protein Symmetry: C1 (View in 3D)

Protein Stoichiometry: A

Macromolecule Content

- Unique protein chains: 2

Display Files ▾   Download Files ▾

2MWY

Mdmx-p53

DOI: 10.2210/pdb2mwy/pdb   **BMRB**: 25377

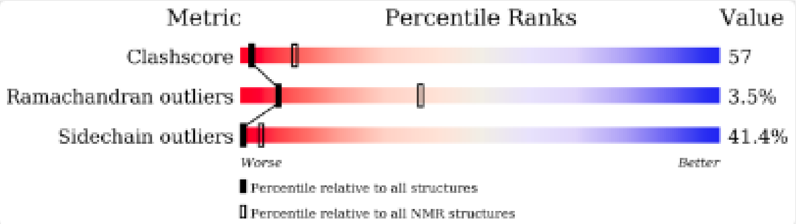
Classification: Cell Cycle / Antitumor Protein
Deposited: 2014-12-03 **Released**: 2016-01-27
Deposition author(s): Grace, C.R.
Organism: Homo sapiens
Expression System: Escherichia coli

Structural Biology Knowledgebase: 2MWY (>12 annotations) SBKB.org

Experimental Data Snapshot

Method: SOLUTION NMR
Conformers Calculated: 100
Conformers Submitted: 20
Selection Criteria: Target Function

wwPDB Validation   Full Report



Literature

Download Primary Citation ▾

Monitoring Ligand-Induced Protein Ordering in Drug Discovery.

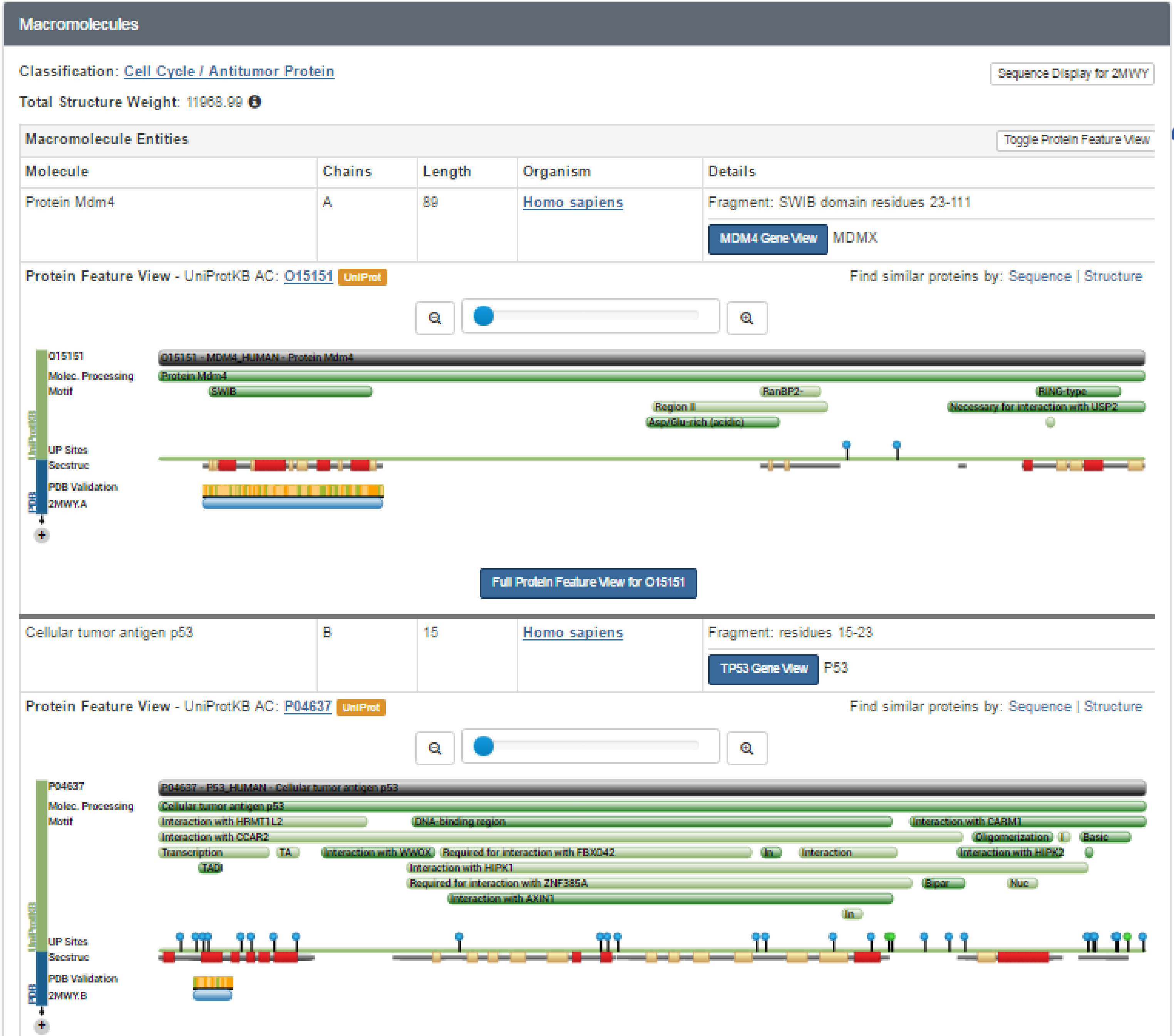
Grace, C.R., Ban, D., Min, J., Mayasundari, A., Min, L., Finch, K.E., Griffiths, L., Bharatham, N., Bashford, D., Kiplin Guy, R., Dyer, M.A., Kriwacki, R.W.

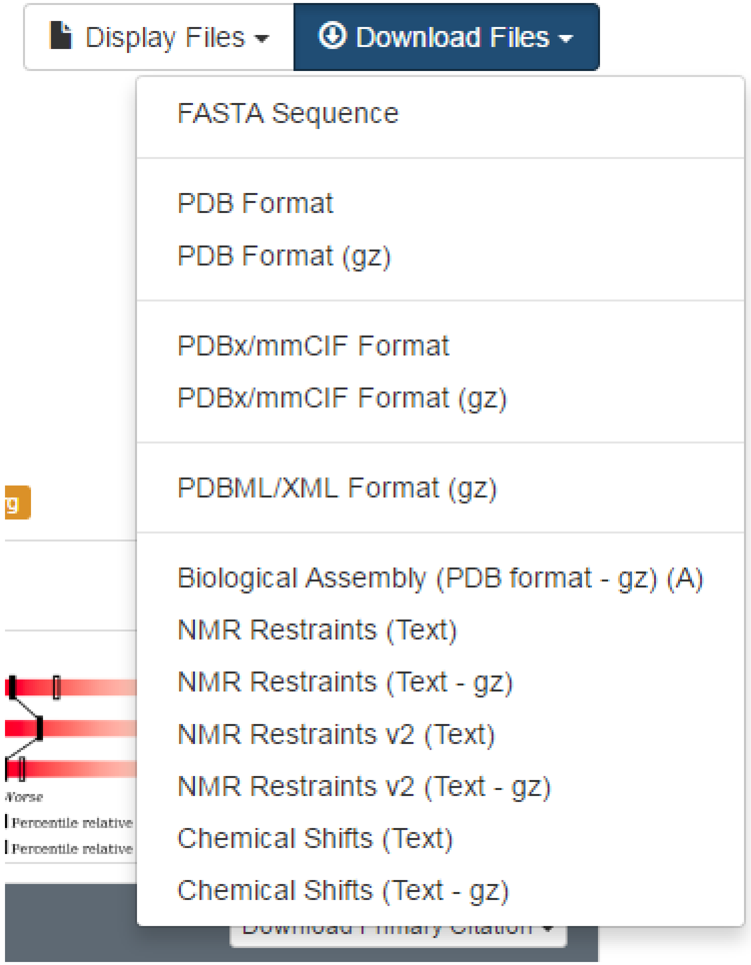
(2016) J.Mol.Biol. **428**: 1290-1303

PubMed: 26812210 Search on PubMed
DOI: 10.1016/j.jmb.2016.01.016

Primary Citation of Related Structures: 2MWY, 2N06, 2N0U, 2N0W, 2N14

- 大分子结果的详细解释：
  - Motif
  - Domain




Display Files
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FASTA Sequence
PDB Format
PDB Format (gz)
PDBx/mmCIF Format
PDBx/mmCIF Format (gz)
PDBML/XML Format (gz)
Biological Assembly (PDB format - gz) (A)
NMR Restraints (Text)
NMR Restraints (Text - gz)
NMR Restraints v2 (Text)
NMR Restraints v2 (Text - gz)
Chemical Shifts (Text)
Chemical Shifts (Text - gz)

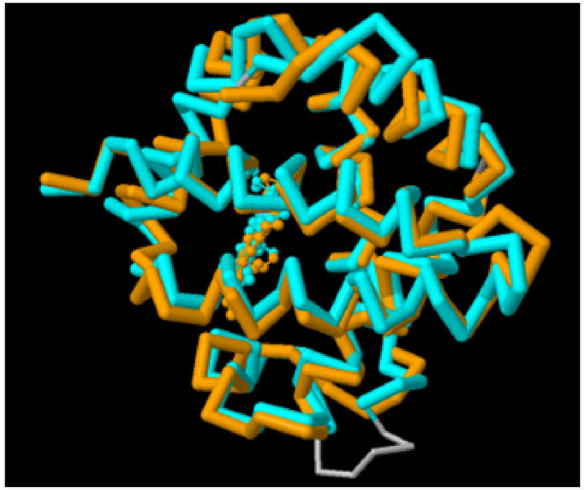
常用的小工具 1

Analyze Options

Sequence & Structure Alignment

RCSB PDB's Comparison Tool calculates pairwise sequence (blast2seq, Needleman-Wunsch, and Smith-Waterman) and structure alignments (FATCAT, CE, Mammoth, TM-Align, TopMatch).

Comparisons can be made for any protein in the PDB archive and for customized or local files not in the PDB. Special features include support for both rigid-body and flexible alignments and detection of circular permutations.



Enter First PDB ID ⟷ Enter Second PDB ID

Select Associated Chain ID

...

Select Associated Chain ID

...

- Select Comparison Method -

Align

More Options

Map Genomic Position to Protein Sequence and 3D Structure

This analysis tool **highlights** the location of a gene position of the human genome on protein sequence and 3D structure.

Note: For looking up gene names, please use the top-bar search (example: BRCA1).

Genome Assembly

GRCh37

Chromosome:

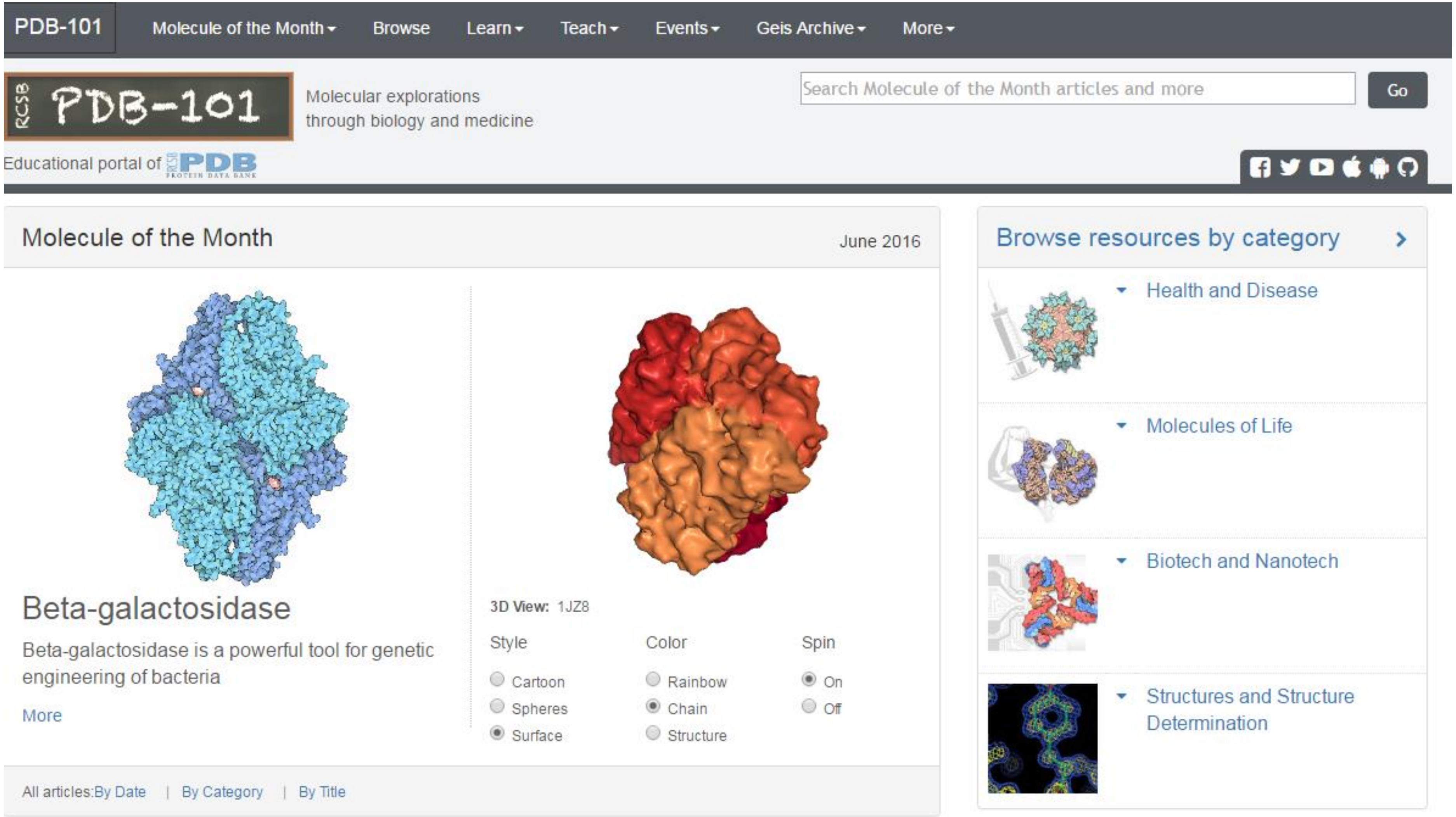
Enter chromosome (example: chr16)

Position:

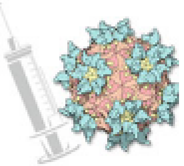
Enter genomic coordinate (example: 226760)

Map Example

PDB-101：非常适合新手的入门篇



Browse resources by category ›



- Health and Disease
  - You and Your Health
  - Immune System
  - HIV and AIDS
  - Diabetes
  - Cancer
  - Viruses
  - Toxins and Poisons
  - Drug Action
  - Drug Resistance

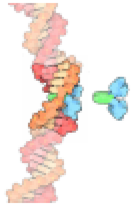
Cancer

Grid List

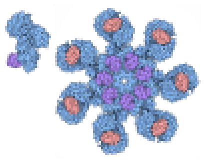
cells growing without controls

Cells in our bodies are carefully regulated so that they divide, grow, and die according to the best plan for keeping us alive. When they lose these controls, they can grow into deadly cancers. Atomic structures have revealed how cell growth is normally controlled, and how cancer cells circumvent these essential controls.

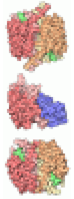
Molecule of the Month Articles (18)



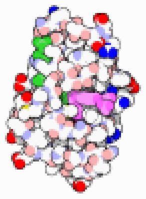
Actinomycin



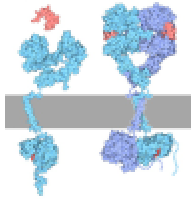
Apoptosomes



Caspases

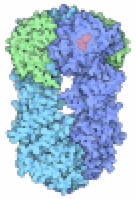


Dihydrofolate Reductase

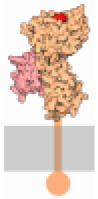


Epidermal Growth Factor

Estrogen Receptor



Hsp90



Major Histocompatibility Complex

总结

- Uniprot 存储了丰富的蛋白序列及结构信息
- PDB存储了丰富的蛋白3D结构信息
- 通过两者的结合，我们可以对蛋白功能进行进一步深入的理解

Thanks